A. MODEL AND TRAINING DETAILS

Our model comprises three components: an audio LM, a symbolic music LM, and a Q-Former connecting the two. This section provides detailed configurations of each component module and the training details.

A.1 Q-Former

The Q-Former is intialized with the MusicBERT-Base model [52]. The added cross-attention layers are randomly initialized. Following Blip-2 [16], we use $K = 32$ learnable queries, each with dimension 768. Symbolic piano arrangements are tokenized in the OctMIDI format [52], which produces note-wise joint embeddings. Overall, the Q-Former comprises 186M parameters, including the learnable queries and note embedding layers.

A.2 Audio LM

We use MusicGen-Large [24] as our audio LM. We discard the text encoder and retain only the music decoder, a 48-layer Transformer. Audio codecs are fed to the decoder and we extract the hidden representations from the 25th layer, as prior probing studies [5, 7–9] suggest that middle layers capture more musically meaningful features. This setup retains 1.7B frozen parameters from MusicGen.

A.3 Symbolic Music LM

For symbolic music arrangement, we adopt MuseCoco-xLarge [3], which is a 24-layer Transformer decoder pre-trained on large-scale symbolic music corpora. We remove its text-related components and keep 1.2B frozen parameters from the music decoder. Symbolic note tokens are converted into the REMI format [53]. Despite the slightly different tokenizations used across stages, we find that the latent representations learned in Stage-I remain compatible with Stage-II. To enable compatibility with MuseCoco, we project $\mathbf{Z}$ into the same embedding dimension as MuseCoco's token embeddings via a linear layer. A LoRa adapter with rank 16 accomodates the added lead sheet condition.

A.4 Training Details

Model training focuses on the 186M Q-Former parameters, which is significantly smaller than the billion-scale LM backbones. In Stage-I, the Q-Former is pre-trained in FP16 using batch size 128 for 10 epochs (130K iteration). The LoRA adaptor in Stage-II adds 5M parameters and we fine-tune the model for another 5 epochs using batch size 32. Both training stages are conducted on four RTX A40 GPUs (48GB each). We use the AdamW optimizer [54] with an initial learning rate of 1e-4, a linear warm-up over the first 1k steps, and a cosine decay schedule to a final rate of 1e-5. At test time, we use top-k sampling with $k = 15$.

B. OBJECTIVE METRICS

We introduce five statistical metrics to evaluate *content preservation* and *style coherence* for the piano cover generation tasks. This section provides the definitions.

B.1 Melody Chroma Accuracy (MCA)

We use the `melody.raw_chroma_accuracy` metric [2] provided by mir_eval [55] to evaluate the similarity between two monophonic melody sequences. For the reference melody, we apply Demucs [56] to isolate the vocal stem from the audio and then extract the F0 contour using pYIN [57] provided by librosa [58]. For the estimated melody, we obtain the melody skyline [59] from the generated piano cover MIDI and convert the MIDI pitches to frequencies. The two melodies are compared position-wise under a tolerance of 50 cents. While this implementation follows [32], we additionally ensure that the two melodies are temporally aligned to the same sequence length.

B.2 Chord Accuracy (CA)

We introduce *Chord Accuracy* from [47, 48] to measure the similarity between two chord sequences. For the reference sequence, when annotated chords are not available, we apply the method of [45] to detect chords from the input audio. For the estimated sequence, we use [60] to detect chords from the generated piano cover MIDI. Both chord sequences are aligned and compared at 1-beat granularity in terms of root and full quality based on the MIREX TeTrads rule. [3]

B.3 Groove Pattern Coherence (GPC)

Grooving Pattern Coherence (*GPC*) evaluates the *grooving pattern* similarity between the generated piano cover and human's arrangement. The *grooving pattern*, which is defined in [49], represents the positions in a MIDI segment at which there is at least one note onset. We consider 4-bar segments at 1/4-beat granularity, deriving *grooving pattern* $\mathbf{g}$ as a 64-dimensional binary vector. The *GPC* over a test piece is defined as follows:

$$GPC = \frac{1}{N}\sum_{n=1}^{N}\cos(\mathbf{g}_n^{\mathrm{hm}}, \mathbf{g}_n^{\mathrm{cv}}), \tag{1}$$

where N is the number of non-overlapping 4-bar segments in the test piece. $\cos(\cdot,\cdot)$ computes the cosine similarity. $\mathbf{g}^{\mathrm{hm}}$ and $\mathbf{g}^{\mathrm{cv}}$ represent the *grooving pattern* feature from the human arrangement and the generated piano cover, respectively. The *GPC* metric is omitted for evaluation on Ballroom/GTZAN, where human's piano arrangement is not available.

B.4 Velocity Contour Coherence (VCC)

Velocity Contour Coherence (*VCC*) evaluates the similarity of the *velocity contour* between the generated piano cover and the human arrangement, using the same formulation as *GPC*. We define the *velocity contour* as a time-series feature representing the average note velocity at each

[2] https://mir-eval.readthedocs.io/latest/api/melody.html #mir_eval.melody.raw_chroma_accuracy

[3] https://mir-eval.readthedocs.io/latest/api/chord.html #mir_eval.chord.tetrads

timestep. It has the same dimensionality and temporal granularity as the *grooving pattern*, but instead of a binary vector, it consists of real values ranging from 0 to 128. The *VCC* metric is omitted for Ballroom/GTZAN, where human's piano arrangement is not available.

B.5 Tempo Accuracy (TA)

Tempo Accuracy (*TA*) evaluates the correctness of the estimated tempo $\mathrm{tp}^{\mathrm{est}}$ relative to the reference (ground-truth) tempo $\mathrm{tp}^{\mathrm{ref}}$. We first define correctness indicator $D(k)$ for one test piece as follows:

$$D(k) = \mathbb{1}\left(\frac{|\mathrm{tp}^{\mathrm{ref}} - k \times \mathrm{tp}^{\mathrm{est}}|}{\mathrm{tp}^{\mathrm{ref}}} < 0.08\right), \qquad (2)$$

where $\mathbb{1}(\cdots)$ is the indicator function. $k \in \{1/2, 1, 2\}$ takes into account the octave ambiguity. The tolerance threshold of 0.08 follows the empirical setting in mir_eval. [4]

We then define tempo accuracy *TA* as:

$$\mathit{TA} = \max\Big(D(1),\ 0.5 \cdot D(1/2),\ 0.5 \cdot D(2)\Big), \qquad (3)$$

where we consider half-tempo and double-tempo matches as partially correct (weight 0.5). This is because tempo perception is known to exhibit octave ambiguity, where perceptions at multiple metrical levels are still considered valid rather than true perceptual errors [61].

We derive the ground-truth tempo from beat annotations when available. On Ballroom/GTZAN, we use the audio tempo estimation results by madmom [44], and we omit the *A2M* baseline in this setting because it already relies on madmom's estimation in its pipeline.

C. SUBJECTIVE EVALUATION DETAILS

Our subjective evaluation is conducted through an online crowdsourcing study, where participants complete a survey consisting of listening and rating tasks. This section provides additional details on the survey design.

C.1 General Instructions

Participants receive the general instructions, which clarify their rights and the conditions of participation:

- Participation is entirely voluntary, and one may withdraw at any time without any negative consequences.
- No personally identifying information is collected; all responses are anonymous and used solely for research purposes.

[4] https://mir-eval.readthedocs.io/latest/api/tempo.html #mir_eval.tempo.detection

C.2 Survey Design

Our survey consists of 6 pages, each presenting 4 versions of piano cover arrangements corresponding to a common test audio piece. The test audio pieces are drawn from the Ballroom and GTZAN datasets and span a variety of genres. The 4 arrangement versions are produced by our model and all baseline models (*PCG2*, *A2M*, and *w/o PT*), respectively. For each model, we select the best result from 3 independently generated samples to avoid occasional low-probability failures (e.g., incomplete or degenerate outputs) from disproportionately affecting subjective evaluation. All models to be evaluated are anonymized, and their orders on each page are randomized.

We acknowledge that participants may use different internal criteria when evaluating music. To minimize such variability, we conduct within-subject ANOVA [50] for data analysis, which ensures that variances in ratings reflect differences among the models rather than participants. To control response quality, we only accept complete evaluation sets; that is, participants have to rate all four models under a common test piece for their responses to be considered valid.

C.3 Completion Time

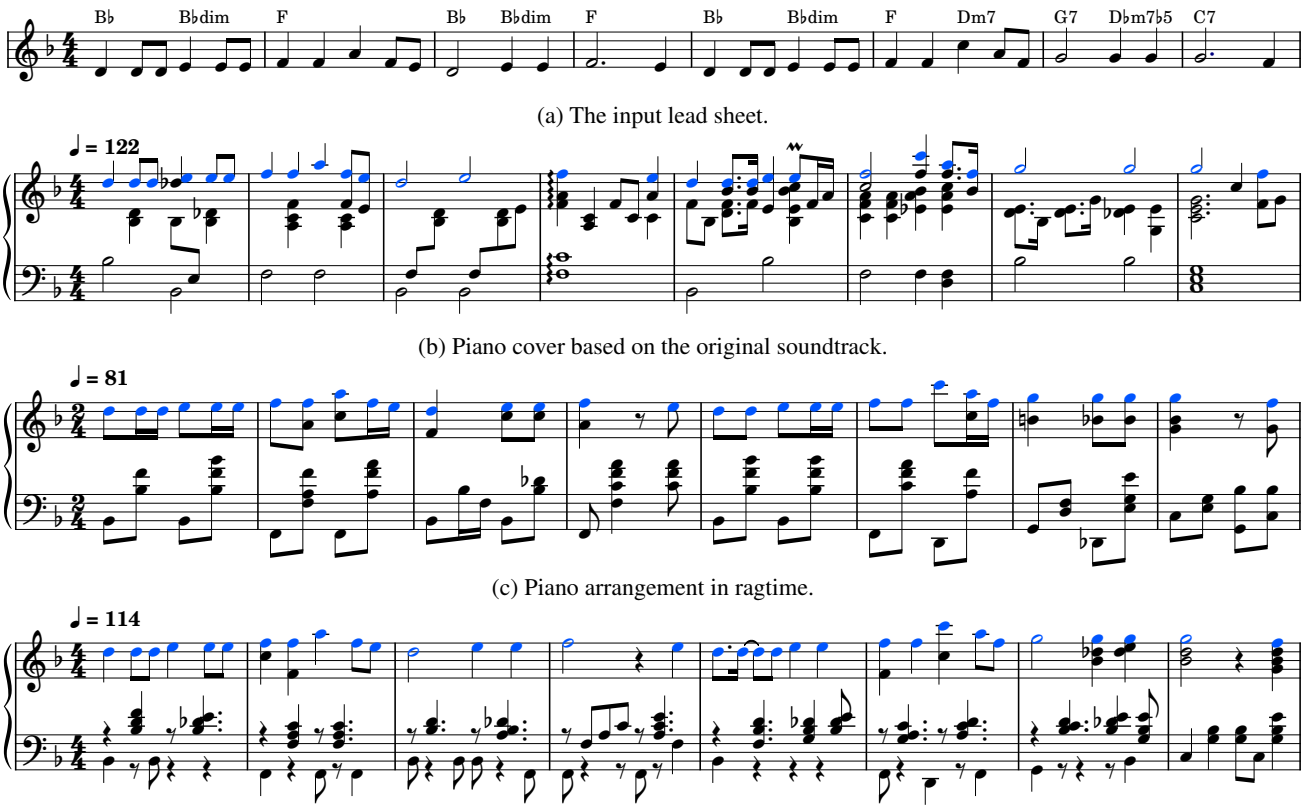
Each participant is randomly assigned 3 out of the 6 pages. On each page, participants first listen to the test audio piece, and then listen to and rate 4 corresponding piano cover samples. All samples in the survey are 16 bars long and rendered to audio using the Cakewalk TTS-1 soundfont, producing approximately 40 seconds of audio per sample. This design targets a total completion time of 10–15 minutes, ensuring that participants have sufficient time to listen carefully without excessive fatigue. The actual completion time observed on average is 12 minutes.

C.4 Participant Profiles

Participants are asked to self-identify their musical background as *amateur*, *intermediate*, or *professional*, following the guidelines below:

- **Amateur**: I enjoy listening to music. I can play/sing/compose short music pieces. I know a little music theory. I can evaluate a composition based on my feelings.
- **Intermediate**: I have some experience in performing, composing, or other music activities. I know a certain amount of music theories that can help me evaluate a composition.
- **Professional**: I am now pursuing/have completed a music degree, or having equivalent background. I am proficient in using music theory to evaluate a composition.

Among the 21 valid responses collected, 6 participants identified as *amateur* (28.6%), 11 as *intermediate* (52.4%), and 4 as *professional* (19.0%). All authors are excluded from the survey.



(a) The input lead sheet.

(b) Piano cover based on the original soundtrack.

(c) Piano arrangement in ragtime.

(d) Piano arrangement in bossa nova.

Figure 4: Audio-to-symbolic arrangement for an 8-bar excerpt from *The Sound of Music*. Score is manually engraved from MIDI for visualisation purpose. Figures 4b to 4d are arranged based on the lead sheet in 4a and an audio reference from the original soundtrack, a ragtime piece, and a bossa nova piece, respectively. Preserved music contents are highlighted in blue note heads. Synthesized audio is provided on the demo page: `https://anonymous55aht.github.io/`.

D. ARRANGEMENT DEMONSTRATION

In this section, we demonstrate the performance of our audio-to-symbolic arrangement model under *freely manipulated* audio style references. Figure 4a shows an 8-bar lead sheet excerpt from the musical *The Sound of Music*. The selected passage features harmonically rich chords, including diminished and seventh chord qualities, which present suitable complexity for arrangement experiments. Figures 4b to 4d showcase the arrangement results conditioned on varied audio references. The 8-bar arrangement is generated using windowed sampling, wherein a 4-bar context window progresses forward every 2 bars and continues sampling conditioned upon the preceding 2 bars.

Figure 4b shows the piano cover from the original *The Sound of Music* soundtrack,[5] which features lush orchestration dominated by string ensembles. Our arrangement captures this orchestral essence through dense, block-chord voicing that emulates the sonority of string sections. Additionally, ornaments such as arpeggios and trills are found to complement the sweeping harmonic textures, which contributes to the music's free-flowing character.

Figure 4c shows an arrangement conditioned on the ragtime classic *The Entertainer*.[6] Following the audio recording, the arrangement's tempo is "not fast," and the piano texture distinctly adopts a ragtime rhythm, with steady bass notes on downbeats and syncopated chordal accents on upbeats.

Figure 4d shows an arrangement conditioned on the bossa nova piece *The Girl from Ipanema*.[7] In this interpretation, the arrangement is characterized by a moderate tempo and distinctive left-hand syncopated patterns characteristic of the bossa nova genre.

Across all three piano arrangements, while distinct music styles are effectively captured from the audio references, the theme melody and harmonic structures remain faithfully preserved. In Figure 4, we highlight melody notes preserved from the lead sheet using blue note heads.

Additional examples are available on our demo page,[8] including arrangements in a wider range of classical, jazz, and pop styles applied to well-known lead sheets, illustrating the flexibility of both style and lead sheet control.

E. LIMITATION

Our proposed method demonstrates the ability to learn implicit music style from audio. At the current stage of this work, we acknowledge that the extracted style

[5] Original audio: `https://youtu.be/6f0T6UV-HiI&t=57`
[6] Ragtime audio: `https://youtu.be/jKlfNfRZL9I&t=11`
[7] Bossa nova audio: `https://youtu.be/DvA_wDOVD10&t=12`
[8] Demo page: `https://anonymous55aht.github.io/`

primarily represents *segment-level global* characteristics. Here, "global" refers to a latent style profile at the level of short 4-bar segments, which is still considerably more fine-grained (and more local) than typical global attributes such as genre labels and textual style tags. Our underlying assumption is that music style remains locally consistent at the segment or bar level. We empirically found that this assumption holds across most music traditions, which guarantees our model to perform reliably under this design. While longer generation can be achieved using windowed sampling, we acknowledge that this approach may smooth over intended stylistic transitions at phrase boundaries, thus leading to diminished expressivity at longer timescales. When considering inter-phrase and longer-term music development, we recognize that a single segment-level style representation is insufficient to capture evolving dynamics. Also, subject to the availability of audio-symbolic data, this work is dedicated to piano arrangement. The cross-modal arrangement of long-term, multi-track music may require hierarchical or temporally adaptive style modelling, which is an important direction for our future work.